

A novel 2nd-order shape function based digital image correlation method for large deformation measurements



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ABSTRACT

Compared with the traditional forward compositional matching strategy, the inverse compositional matching strategy has almost the same accuracy, but has an obviously higher efficiency than the former in digital image correlation (DIC) algorithms. Based on the inverse compositional matching strategy and the auxiliary displacement functions, a more accurate inverse compositional Gauss-Newton (IC-GN2) algorithm with a new second-order shape operator is proposed for nonuniform and large deformation measurements. A theoretical deduction showed that the new proposed second-order shape operator is invertible and can steadily attain second-order precision. The result of the numerical simulation showed that the matching accuracy of the new IC-GN2 algorithm is the same as that of the forward compositional Gauss-Newton (FC-GN2) algorithm and is relatively better than in IC-GN2 algorithm. Finally, a rubber tension experiment with a large deformation of 27% was performed to validate the feasibility of the proposed algorithm.

1. Introduction

A highly efficient and accurate measurement of strain and deformation is very important for studying the mechanical properties and inherent deformation law of different materials and has been a hot topic in the field of mechanics and materials science research. In recent years, as a mature and effective strain measurement technique, two-dimensional digital image correlation (2D-DIC) has many measurement advantages such as high accuracy, noncontact, full-field, three-dimensional, dynamic measurement, low demands on experiment environment, and simple specimen surface treatment [1,2]. Moreover, 3D-DIC was developed by Sutton et al. [3] for out-of-plane deformation measurements. Shao et al. [4] applied 3D-DIC to real-time human pulse monitoring in medical measurements. As a core in DIC techniques, the local matching algorithm between images before and after deformation is always the focus of attention.

At present, the image local matching algorithm mainly includes the feature matching algorithm [1], the phase matching algorithm [5], and the region matching algorithm [6]. These matching algorithms all depend on the interpolation precision. Schreier et al. [7] proved that the use of higher order image interpolation function can significantly improve the searching precision of DIC. Su et al. [8–10] and Gao et al. [11] explored the influences depending on different interpolation methods and different degrees of noise, and they used mean bias error and standard deviation error to represent the systematic error and the

random error. In the matching effect, the feature matching algorithm can only match the specifically required image features, and the matching result is sparse rather than wholly continuous [1]. However, the phase matching algorithm easily falls into the phase singularity and is not suitable for whole matching [5]. By calculating the correlation coefficients of the subset gray levels between the reference image and the target image, the region matching algorithm can obtain dense matching results [6]. In 2D-DIC and 3D-DIC, the region matching algorithm has been widely accepted and developed rapidly.

The region matching algorithm uses a preset correlation function as the optimization objective to find the best matching position. In the matching process, the appropriate subset shape function is selected according to the deformation characteristics of the region of interest (ROI), and the parameters of the subset shape function are used to describe the deformation characteristics of the local coordinate system [1]. Similar to the shape function in the finite element method (FEM), the higher order shape function can be used to describe more complicated mechanical deformation, but it will also require more computations. Usually, the first-order shape function is suitable for small deformation measurements, and can effectively characterize the translation, rotation, and uniform deformation. The second-order shape function can effectively reflect the complex deformation, thus making the nonuniform and large deformation measurement feasible [12]. Lu et al. [13] proved that the second-order shape function has

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better precision than the first-order one and has been successfully applied to the large deformation measurement of materials. Wang et al. [14] and Wang et al. [15] have investigated the influence of data noise on the matching accuracy, which can be reduced by the more accurate image interpolation and the selection of the appropriate correlation function. Schreier et al. [16] and Xu et al. [17] have respectively researched the systematic errors arising from the shape functions and subset sizes, and they found a larger subset size would probably cause a more significant errors.

The region matching algorithm is a complicated nonlinear optimization problem and is generally optimally solved by the Newton-Raphson iteration method or trust-domain iteration method. The commonly used iterative updating methods are the forward additive Gauss-Newton (FA-GN) algorithm and the forward compositional Gauss-Newton (FC-GN) algorithm, both of which have been proved to be equivalent to each other by Shum and Szeliski [18]. The traditional FC-GN algorithm is described as follows. The deformation of the target subset is realized by the shape function and is searched by the maximum correlation with the reference subset in the iteration. The subpixel interpolation of the target image for each iteration is completed to obtain the partial derivatives, the Jacobian matrix and its inverse matrix calculation, which consumes most of the computational time in the subpixel matching process.

Baker and Matthews [19] found that the inverse compositional Gauss-Newton (IC-GN) algorithm can effectively reduce the calculation time because in each matching process the image partial derivative, the Jacobian matrix and its inverse matrix are calculated only once until the matching is accomplished. The Jacobian determinant is closed to 1 when the FC-GN form is transformed to its inverse form, the IC-GN algorithm; therefore the two algorithms are equivalent. On the basis of the IC-GN algorithm, Pan et al. [20,21] added the zero-normalized sum of squared differences (ZNSSD) correlation function and a seed parallel searching strategy. Shao et al. [22] proposed an anti-noised IC-GN algorithm using the SSD correlation function. Gao et al. [23] suggested that a second-order inverse compositional Gauss-Newton (IC-GN2) algorithm be applied to 3D-DIC matching and validated using simulated speckle images. In addition, the second-order shape function was applied to the IC-GN2 algorithm for out-of-plane deformation [24], 3D deformation of satellite antenna surface [25], and civil engineering materials [26,27].

However, Gao's IC-GN2 algorithm has an imperfect second-order shape operator. After a brief introduction of the FA-GN, FC-GN, and IC-GN algorithms, the IC-GN2 algorithm with a new second-order shape operator is proposed in this paper. Theoretical error analysis showed that the new IC-GN2 algorithm can achieve a stable second-order accuracy. The simulated speckle numerical comparison and the rubber tension experiment for large deformation measurement verified the validity of the IC-GN2 algorithm with the proposed second-order shape operator.

2. Fundamental principles

2.1. FA-GN algorithm

The displacement shape function plays an important role in the traditional FA-GN algorithm, since it describes the displacement mode of a point. Therefore, the deformation relationship between the target subset and the reference subset can be established. The first-order displacement shape operator can be written as

$$W(\mathbf{x}, p) = \begin{bmatrix} 1 & 0 & 0 \\ u & u_x + 1 & u_y \\ v & v_x & v_y + 1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix} = \mathbf{A}(p)\mathbf{x}, \quad (1)$$

where $\mathbf{x} = [1 \ x \ y]^T$ is the polar coordinate of the reference image under the local coordinate system. $p = [u \ u_x \ u_y \ v \ v_x \ v_y]^T$ is the vector

concluding first-order deformation parameters of the target image and the subscripts denote the partial derivatives at the subset center. $\mathbf{A}(p)$ is a linear transformation matrix with size of 3×3 for the first-order deformation vector p . In the DIC process, the first-order deformation vector p is obtained through minimizing the image correlation function as

$$C(p) = \sum_x [f(\mathbf{x}) - g(W(\mathbf{x}, p))]^2, \quad (2)$$

where $f(\mathbf{x})$ and $g(W(\mathbf{x}, p))$ denote the gray matrices in the reference subset and the target subset, respectively. Every iteration in the FA-GN algorithm aims to minimize the following deviation function as

$$\begin{aligned} & \sum_x [f(\mathbf{x}) - g(W(\mathbf{x}, p + \Delta p))]^2 \\ & \approx \sum_x \left[g(W(\mathbf{x}, p)) + \nabla g(W(\mathbf{x}, p)) \frac{\partial W(\mathbf{x}, p)}{\partial p} \Delta p - f(\mathbf{x}) \right]^2. \end{aligned} \quad (3)$$

After the deformation increment Δp is obtained, the deformation parameter is updated by $p \leftarrow p + \Delta p$.

2.2. FC-GN algorithm

The FC-GN was first proposed by Shum and Szeliski [18]. The deformation parameter p is indirectly updated by the compositional form $W(W(\cdot, \Delta p), p)$ of the shape operator $W(\cdot, p)$ in the matching process, instead of being directly updated by the deformation parameter p . The increment Δp for each iteration aims to minimize the following deviation function as

$$\begin{aligned} & \sum_x [f(\mathbf{x}) - g(W(W(\mathbf{x}, \Delta p), p))]^2 \\ & \approx \sum_x \left[g(W(W(\mathbf{x}, 0), p)) + \nabla g(W(W(\mathbf{x}, 0), p)) \frac{\partial W(W(\mathbf{x}, 0), p)}{\partial W(\mathbf{x}, 0)} \frac{\partial W(\mathbf{x}, 0)}{\partial p} \Delta p - f(\mathbf{x}) \right]^2 \\ & = \sum_x \left[g(W(\mathbf{x}, p)) + \nabla g(W(\mathbf{x}, p)) \frac{\partial W(\mathbf{x}, p)}{\partial \mathbf{x}} \frac{\partial W(\mathbf{x}, 0)}{\partial p} \Delta p - f(\mathbf{x}) \right]^2. \end{aligned} \quad (4)$$

Instead of directly updating the parameter p , this strategy updates the shape operator as

$$W(\mathbf{x}, p) \leftarrow W(W(\mathbf{x}, \Delta p), p) \quad (5)$$

Comparing Eq. (5) with Eq. (3), we can see that the FC-GN algorithm needs to compute the partial derivatives of $\frac{\partial W(\mathbf{x}, 0)}{\partial \mathbf{x}}$ and $\frac{\partial W(\mathbf{x}, 0)}{\partial p}$, while the FA-GN algorithm only computes the partial derivative of $\frac{\partial W(\mathbf{x}, 0)}{\partial p}$. The above two iterations are equivalent to the following updating process as

$$\text{FA - GN updating: } W(\mathbf{x}, p + \Delta p) \approx W(\mathbf{x}, p) + \frac{\partial W}{\partial p} \Delta p \quad (6-1)$$

$$\begin{aligned} \text{FC - GN updating: } & W(W(\mathbf{x}, \Delta p), p) \approx W(\mathbf{x} + \frac{\partial W}{\partial p} \Delta p, p) \approx W(\mathbf{x}, \\ & p) + \frac{\partial W}{\partial \mathbf{x}} \frac{\partial W}{\partial p} \Delta p \end{aligned} \quad (6-2)$$

2.3. IC-GN algorithm

The IC-GN algorithm was first proposed by Baker et al. [19], which is a modified version of the FC-GN algorithm. It minimizes the following deviation function as

$$\begin{aligned}
& \sum_x [f(W(\mathbf{x}, p)) - g(W(\mathbf{x}, p))]^2 \\
& \approx \sum_x \left[f(W(\mathbf{x}, 0)) + \nabla f(W(\mathbf{x}, 0)) \frac{\partial W}{\partial p} \Delta p - g(W(\mathbf{x}, p)) \right]^2 \\
& = \sum_x \left[f(\mathbf{x}) + \nabla f(\mathbf{x}) \frac{\partial W}{\partial p} \Delta p - g(W(\mathbf{x}, p)) \right]^2.
\end{aligned} \quad (7)$$

The IC-GN algorithm directly applies the small shape operator $W(\mathbf{x}, p)$ to the reference subset so as to obtain a higher correlation coefficient, and then this operator is reversely applied to the target subset to update the shape operator $W(\mathbf{x}, p)$ as

$$W(\mathbf{x}, p) \leftarrow W(W^{-1}(\mathbf{x}, \Delta p), p). \quad (8)$$

In Eq. (1), if the linear transformation matrix $A(p)$ is nonsingular, it is easily known that the operator $W(\cdot, p)$ is an invertible in the polar coordinates $\mathbf{x}$. Thus, the operator $W(W^{-1}(\mathbf{x}, \Delta p), p)$ can be written as

$$W(W^{-1}(\mathbf{x}, \Delta p), p) = A(p)A^{-1}(\Delta p)\mathbf{x} \quad (9)$$

where

$$A(\Delta p) = \begin{bmatrix} 1 & 0 & 0 \\ \Delta u & \Delta u_x + 1 & \Delta u_y \\ \Delta v & \Delta v_x & \Delta v_y + 1 \end{bmatrix} \quad (10)$$

The FC-GN and the IC-GN algorithms have an equivalent effect, but the latter is more efficient. In the IC-GN algorithm, the partial derivative $\nabla f(\mathbf{x})$ of the reference subset in Eq. (7) remains unchanged during the iterative process. When the partial derivative $\nabla f(\mathbf{x})$ in the first iteration is calculated to construct the Jacobian matrix $\mathbf{J}$, the matrix $(\mathbf{J}^T \mathbf{J})^{-1} \mathbf{J}^T$ is constant and can save computation in the Newton-Raphson iteration process. In the FC-GN algorithm, however, the partial derivative $\nabla g(W(\mathbf{x}, p))$ in Eq. (4) needs to be repeatedly calculated by image interpolation operation in each iteration, and it will waste a large amount of computation. Sutton et al. [12] through a large number of experiments proved that the IC-GN algorithm has better speed and accuracy than the FC-GN algorithm.

2.4. The new IC-GN algorithm

Similar to the FEM, the higher order displacement shape function can, with better accuracy, estimate the more complicated deformation mode of the deformed subset. In the FA-GN2 algorithm, the second-order Taylor expansion is used to construct the second-order shape operator as

$$W(\mathbf{x}, p) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ u & u_x + 1 & u_y & \frac{1}{2}u_{xx} & u_{xy} & \frac{1}{2}u_{yy} \\ v & v_x & v_y + 1 & \frac{1}{2}v_{xx} & v_{xy} & \frac{1}{2}v_{yy} \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \\ x^2 \\ xy \\ y^2 \end{bmatrix} = \mathbf{E}(p)\mathbf{x}, \quad (11)$$

where $\mathbf{x} = [1 \ x \ y \ x^2 \ xy \ y^2]^T$ is second-order form of the polar coordinate in the reference image and the corresponding deformation parameter is $p = [u \ u_x \ u_y \ u_{xx} \ u_{xy} \ u_{yy} \ v \ v_x \ v_y \ v_{xx} \ v_{xy} \ v_{yy}]^T$. The second-order linear transformation matrix $\mathbf{E}(p)$ is a 3×6 matrix, which is composed of two 3×3 matrices of $\mathbf{A}(p)$ and $\mathbf{B}(p)$, namely

$$\mathbf{E}(p)_{3 \times 6} = [\mathbf{A}(p)_{3 \times 3} \ \mathbf{B}(p)_{3 \times 3}] \quad (12)$$

and

$$\mathbf{B}(p) = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{2}u_{xx} & u_{xy} & \frac{1}{2}u_{yy} \\ \frac{1}{2}v_{xx} & v_{xy} & \frac{1}{2}v_{yy} \end{bmatrix}. \quad (13)$$

In the Newton-Raphson iteration, the second-order shape operator can make the target subset present more abundant nonuniform deformation with curve shape edges. As shown in Fig. 1, the square

subset of 41×41 pixels presents different deformation shapes under the first-order and second-order shape operators after three Newton-Raphson iterations, respectively.

It should be noted that the second-order shape operator (Eq. (11)) is suitable for the FA-GN2 algorithm and the FC-GN2 algorithm, but it cannot be used in the IC-GN2 algorithm. This is because the second-order shape operator $W(\cdot, p)$ is not invertible. Therefore, the matrix $\mathbf{E}(p)$ in Eq. (11) is not a square matrix and can not be directly applied to the IC-GN2 algorithm.

To solve this problem, it is necessary that $W(\cdot, p)$ must be an invertible linear operator and that its corresponding linear transformation matrix $\mathbf{E}(p)$ must be a square matrix. Enlarging $\mathbf{E}(p)$ in Eq. (11) from a 6×3 matrix to a 6×6 matrix is a direct way to achieve this aim and Gao et al. [23] proposed a second-order shape operator for the IC-GN2 algorithm as

$$W(\mathbf{x}, p) = \mathbf{G}(p)\mathbf{x} \quad (14)$$

and

$$\mathbf{G}(p)_{6 \times 6} = \begin{bmatrix} \mathbf{E}(p)_{3 \times 6} \\ \mathbf{F}(p)_{3 \times 6} \end{bmatrix} = \begin{bmatrix} \mathbf{A}(p)_{3 \times 3} & \mathbf{B}(p)_{3 \times 3} \\ \mathbf{C}(p)_{3 \times 3} & \mathbf{D}(p)_{3 \times 3} \end{bmatrix}, \quad (15)$$

where a 3×6 linear transformation matrix $\mathbf{F}(p)$ is composed of two 3×3 matrices as

$$\mathbf{F}(p)_{3 \times 6} = [\mathbf{C}(p)_{3 \times 3} \ \mathbf{D}(p)_{3 \times 3}] \quad (16)$$

and

$$\mathbf{C}(p) = \begin{bmatrix} u^2 & 2uu_x + 2u & 2uu_y \\ uv & u_x v + uv_x + v & u_y v + uv_y + v \\ v^2 & 2vv_x & 2vv_y + 2v \end{bmatrix}, \quad (17)$$

$$\mathbf{D}(p) = \begin{bmatrix} 2u_x^2 + uu_{xx} + 2u_x & 2uu_{xy} + 2u_x u_y + 2u_y & u_y^2 + uu_{yy} \\ +1 & & \\ \frac{1}{2}u_{xx}v + u_x v_x & u_{xy}v + uv_{xy} + u_x v_y + u_y v_x & \frac{1}{2}u_{yy}v + u_y v_y \\ + \frac{1}{2}uv_{xx} + v_x & + u_x + v_y + 1 & + \frac{1}{2}uv_{yy} + u_y \\ v_x^2 + vv_{xx} & 2vv_{xy} + 2v_x v_y + 2v_y & 2v_y^2 + vv_{yy} + 2v_y \\ & & +1 \end{bmatrix}. \quad (18)$$

The enlarged square matrix $\mathbf{G}(p)_{6 \times 6}$ is invertible and can be used for the IC-GN2 algorithm. In this paper, a more rigorous second-order shape operator, the reversible linear operator $W(\cdot, p)$ can be expressed as

$$W(\mathbf{x}, p) = \mathbf{G}'(p)\mathbf{x} + \mathbf{o}(x^2 + y^2) \quad (19)$$

Compared with Gao's shape operator (Eq. (14)), the newly constructed operator has a more stable accuracy reaching $\mathbf{o}(x^2 + y^2)$. According to the derivation in the Appendix, the new 6×6 square linear transformation matrix can be rewritten as

$$\mathbf{G}'(p)_{6 \times 6} = \begin{bmatrix} \mathbf{E}(p)_{3 \times 6} \\ \mathbf{F}'(p)_{3 \times 6} \end{bmatrix} = \begin{bmatrix} \mathbf{A}(p)_{3 \times 3} & \mathbf{B}(p)_{3 \times 3} \\ \mathbf{C}'(p)_{3 \times 3} & \mathbf{D}'(p)_{3 \times 3} \end{bmatrix}. \quad (20)$$

Compared with the Gao's square linear transformation matrix (Eq. (15)), two 3×3 second-order transformation matrices in Eq. (20) are different, which lead to a higher precision, as shown in the following equations:

$$\mathbf{C}'(p) = \mathbf{C}(p) + \begin{bmatrix} 0 & 2u & 0 \\ 0 & v & u \\ 0 & 0 & 2v \end{bmatrix} \quad (21)$$

and

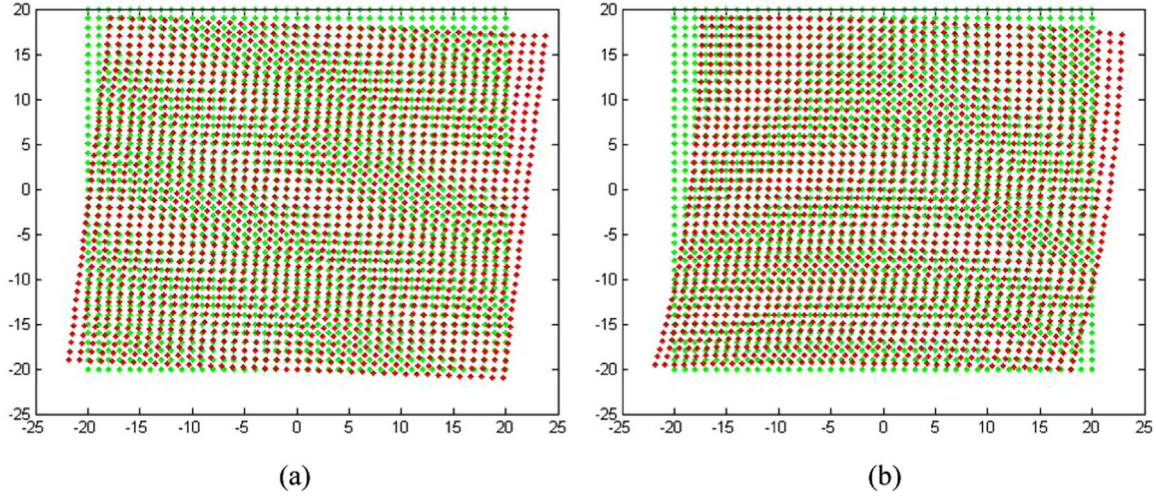


Fig. 1. Deformation shapes under (a) the first-order shape operator and (b) the second-order shape operator.

$$\mathbf{D}'(p) = \mathbf{D}(p) + \begin{bmatrix} 2u_x & 2u_y & 0 \\ v_x & u_x + v_y & u_y \\ 0 & 2v_x & 2v_y \end{bmatrix}. \quad (22)$$

The newly constructed invertible second-order shape operator $W(\mathbf{x}, p)$ can be updated in the IC-GN2 algorithm as

$$W(\mathbf{x}, p) \leftarrow W(W^{-1}(\mathbf{x}, \Delta p), p) = B(p)B^{-1}(\Delta p)\mathbf{x} \quad (23)$$

If $U(x, y)$ and $V(x, y)$ meet the second-order displacement mode between the reference subset and the target subset, namely

$$\begin{cases} x' - x = U(x, y) + o(x^2 + y^2) \\ y' - y = V(x, y) + o(x^2 + y^2) \end{cases} \quad (24)$$

knowing that $|x + y| \leq \sqrt{2}(x^2 + y^2)^{\frac{1}{2}}$, there is

$$\begin{aligned} x'y' &= (x + U)(y + V) + (U + V + x + y)o(x^2 + y^2) + o((x^2 + y^2)^2) \\ &= (x + U)(y + V) + (U + V)o(x^2 + y^2) + o((x^2 + y^2)^{\frac{3}{2}}) \end{aligned} \quad (25)$$

Expanding Eq. (11) of the FA-GN2 operator as

$$\begin{cases} U(x, y) = u + u_x x + u_y y + \frac{1}{2}u_{xx}x^2 + u_{xy}xy + \frac{1}{2}u_{yy}y^2 \\ V(x, y) = v + v_x x + v_y y + \frac{1}{2}v_{xx}x^2 + v_{xy}xy + \frac{1}{2}v_{yy}y^2 \end{cases} \quad (26)$$

By ignoring high-order terms like $x^3, y^3, x^2y, xy^2, x^3y, x^2y^2, xy^3, x^4$ and y^4 , the $x'y'$ in Eq. (25) can be seen as the following form (same to Gao's shape operator in Eq. (14)):

$$\begin{aligned} x'y' &= uv + (u_x v + uv_x + v)x + (u_y v + uv_y + u)y \\ &\quad + \left(\frac{1}{2}u_{xx}v + u_x v_x + \frac{1}{2}uv_{xx} + v_x \right)x^2 + (u_{xy}v + uv_{xy} + u_x v_y + u_y v_x)xy \\ &\quad + \left(\frac{1}{2}u_{yy}v + u_y v_y + \frac{1}{2}uv_{yy} + u_y \right)y^2 \end{aligned} \quad (27)$$

The above Eq. (27) may lose second-order accuracy because of ignoring too many high-order terms. It is seen from Eq. (25) that $U(x, y) + (x, y)$ can be regarded as a bounded function if the deformation is small enough, then there is $x'y' = (x + U)(y + V) + o(x^2 + y^2)$. It means that $x'y' = (x + U)(y + V)$ has a second-order accuracy, which was actually used in Gao's shape operator, and it is reasonable to ignore these terms under the condition of bounded function. But if the deformation is relatively large or the local deformation is non-uniform, $U + V$ may have the same order as $(x^2 + y^2)^q$ in certain positions, where the precision of $x'y' = (x + U)(y + V)$ is lower than the second-order of $o(x^2 + y^2)$ when there is $q < 0$. Therefore, Gao's shape operator (Eq. (15)) will probably lose the second-order precision in the case of local non-uniform deformation when there is $q < 0$. In contrast, the proposed shape

operator (Eq. (20)) is capable of stably achieving the second-order precision in the case of large deformation or local non-uniform deformation.

3. Numerical verifications

Numerical simulation was used to evaluate the accuracy and error of the proposed IC-GN algorithm with the new second-order shape operator. The speckle simulation method proposed by Zhou et al. [26] has proved to be a convenient implementation and has high accuracy for theoretical speckle generation. There are K Gaussian light spots simulated and projected onto the plane for the theoretical speckle generation, and the light intensity $I(\mathbf{x})$ at coordinate $\mathbf{x}$ is calculated by the superposition of all Gaussian light spots as

$$I(\mathbf{x}) = \sum_{k=1}^K I_k e^{-\frac{\|\mathbf{x} - \mathbf{x}_k\|^2}{r^2}}, \quad (28)$$

where I_k is the light intensity of the k th Gaussian light spot, and r is the influence radius of the light-spot. $\mathbf{x}_k$ is the coordinate of the k th Gaussian light spot, which is randomly generated by the computer.

If the deformation $\mathbf{x}' = P(\mathbf{x})$ is added to the simulated speckle image, where P is the shape operator, then the light intensity of the deformed speckle image is written as

$$I'(\mathbf{x}) = I(P^{-1}(\mathbf{x}')) = \sum_{k=1}^K I_k e^{-\frac{\|P^{-1}(\mathbf{x}') - \mathbf{x}_k\|^2}{r^2}} \quad (29)$$

A discrete sampling procedure is needed to generate a grayscale image with a size of $M \times N$. To obtain the uniform illumination distribution, we used the Hamersley sampling method [27] with low-discrepancy sequence in the plane to select the coordinates $\mathbf{x}_k$ ($k=1, 2, \dots, K$).

In the first simulation example, the new operator's performance in large uniform deformation will be presented, in which the number of Gaussian light spot was $K=20,000$ and the spot radius was $r=1.5$ pixels. The simulated speckle image had a size of 400×400 pixels by sampling along the x -axis and y -axis with a step of 1 pixel, as shown in Fig. 2(a). A first-order shape operator with the predesigned linear polynomial deformation is written as

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} u \\ v \end{pmatrix}, \quad (30)$$

where the deformation parameters for the first-order shape operator are selected as $p_{11}=-0.02$, $p_{12}=0.05$, $p_{21}=-0.01$, and $p_{22}=-0.1$ and the rigid body displacements are $u=v=0$. The deformed speckle image was

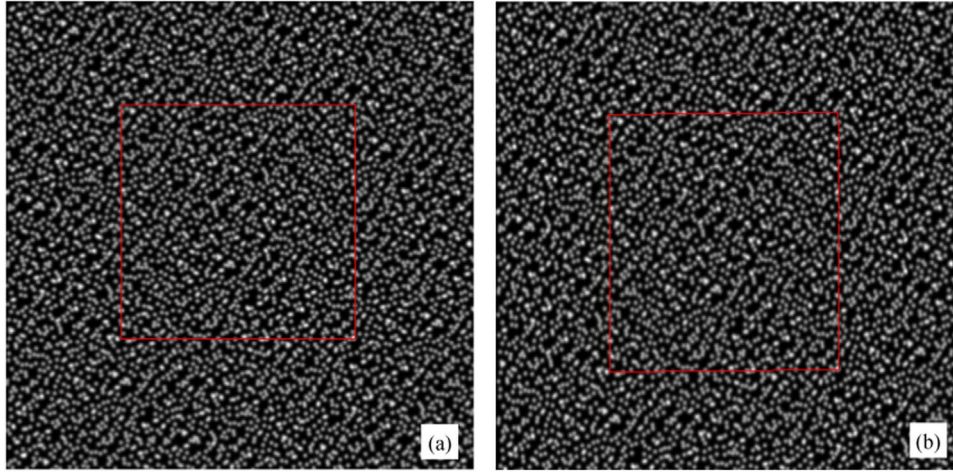


Fig. 2. Simulated speckle images (a) before and (b) after the deformation.

simulated, as shown in Fig. 2(b).

The selection of the correlation function, as the target function for optimization iterations, is very important for the DIC matching. The ZNSSD was chosen as the correlation function, which is insensitive to the data noises and the linear transform of the subset gray in the target image [12]. The ZNSSD correlation function is written as

$$C_{ZNSSD}(p) = \sum_{x=-M}^M \sum_{y=-M}^M \left[\frac{f(x, y) - \bar{f}}{\sqrt{\sum_{x=-M}^M \sum_{y=-M}^M [f(x, y) - \bar{f}]^2}} - \frac{g(x', y') - \bar{g}}{\sqrt{\sum_{x=-M}^M \sum_{y=-M}^M [g(x', y') - \bar{g}]^2}} \right]^2 \quad (31)$$

where the subset size is $(2M+1) \times (2M+1)$. $\bar{f}$ and $\bar{g}$ are the mean gray values for the reference subset and target subset, respectively. The cubic B-spline interpolation was used to calculate the gray value and partial derivatives at subpixel points.

Correlation matching calculations of the simulated speckle images were performed after three iterations by different second-order DIC algorithms: FC-GN2, Gao's IC-GN2 and the proposed IC-GN2. The average error increased with the subset size for different algorithms, as shown in Fig. 3. When the subset size was large enough (larger than about 43 pixels), the proposed IC-GN2 algorithm showed more stable matching precision. It is recommended to select the subset size from 43 pixels to 61 pixels in practical applications.

In the second simulation example, a sinusoidal deformation was

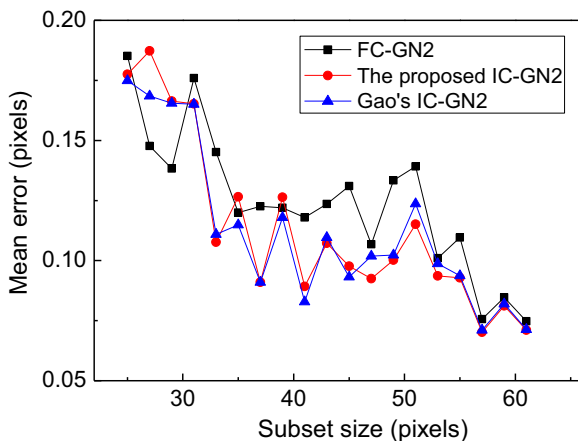


Fig. 3. Average errors of the three second-order DIC algorithms after three iterations under different subset sizes.

conducted to verify the capability of these second-order operators in the non-uniform deformation. The priori deformation was set as the following form

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = P(x, y) = \begin{pmatrix} x + sT \sin(\frac{2\pi}{T}(x-x_0)) \\ y \end{pmatrix}, \quad (32)$$

where x_0 is the x coordinate of center point in the simulated images, the deformation parameters of $T=250$ and $s=0.001-0.01$ with a step size=0.001. The simulated speckle images had a size of $M \times N=200 \times 200$ pixels, in which the number of Gaussian light spot was $K=5000$ and the spot radius was $r=1$ pixel. Because the x -strain is about $\epsilon_x = 2\pi s \cos(\frac{2\pi}{T}(x-x_0))$, the priori deformation can be viewed as a non-uniform one, and it is ranged form $0-2\pi s$. In deformation matching process, the subset size was set to 33 pixels. And here, the expectation of error of these second-order operators is defined as

$$e = \frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N [u(x, y) - u_0(x, y)], \quad (33)$$

where $u(x, y)$ is the measured x -displacement of the point (x, y) and $u_0(x, y)$ is the real x -displacement. As shown in Fig. 4(a), the expectation of error shows a waving trend responding to the parameter s , and the half-wave length of the fluctuations seems to enlarge with increasing of s . The proposed IC-GN2 has a relatively better performance than Gao's IC-GN2 in the sinusoidal deformation.

In the third simulation example, a quadratic deformation was also investigated as the following form:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = P(x, y) = \begin{pmatrix} x + sx^2 \\ y \end{pmatrix}, \quad (34)$$

where the tested deformation parameter a was ranged form 0 to 0.0002 with a stepsize 0.00001. Since the x -strain is $\epsilon_x = 2sx$, $a=200 \times 200$ pixels image has a maximum strain of 400s. But under the quadratic deformation, there was no significant difference between the two shape operators as shown in Fig. 4(b). The expectation of error shows the two IC-GN2 algorithms are better than the FC-GN2 and the proposed IC-GN2 has a relatively better performance than Gao's IC-GN2.

4. Large deformation experiment of rubber tension

A large tension deformation experiment of rubber was performed to check the performance of three second-order DIC algorithms. The rubber specimen with a size of 25 mm $\times$ 100 mm $\times$ 3 mm was sprayed with random white and black speckles, as shown in Fig. 5. The speckle images after the rubber plane deformation were acquired by a CCD camera (Guppy F080b) with resolution of 1024 $\times$ 768 pixels. The frame

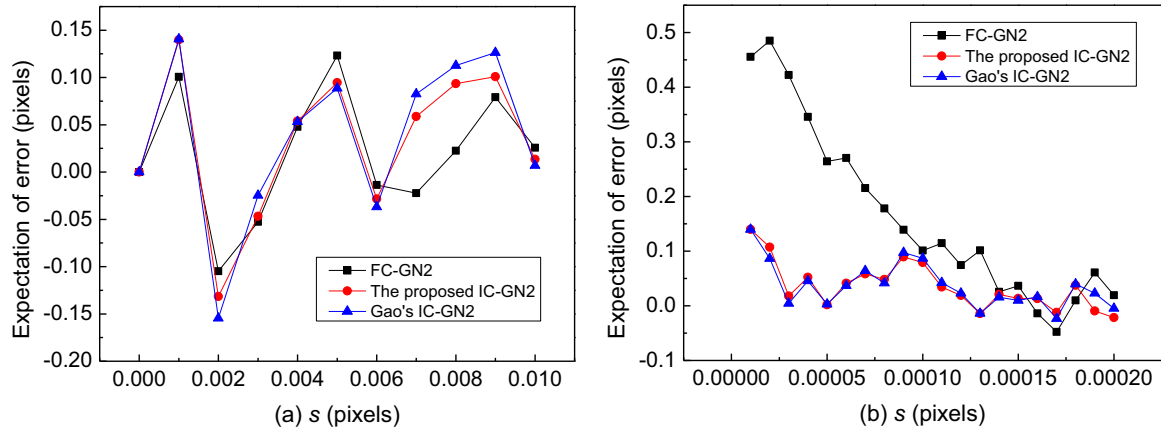


Fig. 4. Expectation of error of three second-order DIC algorithms after three iterations under (a) sinusoidal and (b) quadratic non-uniform deformations.

acquisition rate was 1 fps and the tensile rate was 1 mm/min.

Fig. 5(a) shows the surface speckle image of the rubber specimen before the deformation, and the tensile direction is along the negative y -axis. Fig. 5(b) is the ROI area with 190×190 pixels (dotted rectangle in Fig. 5(a)). The grid step for the correlation calculation was 10 pixels, so there was a total of 400 calculated points.

The subset size was selected as 45×45 pixels, and the cubic B-spline interpolation was applied to calculate the gray value and partial derivatives at subpixel points. For the large deformation of about 7.5%, the displacement fields of ROI were calculated by the proposed IC-GN2, as shown in Fig. 6. It can be seen that there is an obvious Poisson contraction effect in the transverse x -direction (Fig. 6(a)) and that a large tensile displacement in the longitudinal y -direction has reached 40 pixels (Fig. 6).

For the large deformation or geometric nonlinear properties, a quadratic polynomial fitting was used to get a continuous displacement function for strain tensor calculation, as expressed in following equation:

$$\varepsilon_{ij} = \frac{1}{2}(u_{i,j} + u_{j,i} + u_{k,i}u_{k,j}). \quad (35)$$

The strain fields of the rubber specimen under about 7.5% tension deformation are shown in Fig. 7.

The correlation coefficient field generated by the matching results is usually an important index for judging the proposed algorithm. The correlation coefficient distribution of the deformation from 7.5% to 27% is given in Fig. 8, which shows that the correlation coefficient

decreased with the deformation. Even if the deformation increases to about 27% (Fig. 8(d)), the minimum correlation coefficient of the proposed algorithm is still more than 0.88.

Because the matrices $C(p)$ and $D(p)$ in the linear transformation matrix $G(p)$ of the Gao's shape operator (Eq. (15)) make an adjustment to the matrices $C'(p)$ and $D'(p)$ in the proposed second-order shape operator (Eq. (20)), the calculation time for these two shape operators is the same as in the IC-GN algorithm. The distributions of the average correlation coefficient with the subset size for three IC-GN2 algorithms are given in Fig. 9.

It can be seen from Fig. 9(a) that the performance of the FC-GN2 algorithm is better than those of the two IC-GN2 algorithms when the deformation was about 7.5%. When the deformation reached 27% (Fig. 9(d)), the performance of Gao's IC-GN2 algorithm was still lower than that of the FC-GN2 algorithm, but the mean correlation coefficient of the proposed IC-GN2 algorithm was close to that of the FC-GN2 algorithm with the increase in the subset size. Especially, the performance of the proposed IC-GN2 algorithm exceeded that of the FC-GN2 algorithm when the subset size was larger than 43 pixels. It proves the proposed IC-GN2 algorithm is effective in large deformation measurements.

5. Conclusions

By introducing auxiliary displacement functions to the second-order shape operator, we were able to propose a new reversible linear transformation matrix for constructing the IC-GN2 algorithm in the

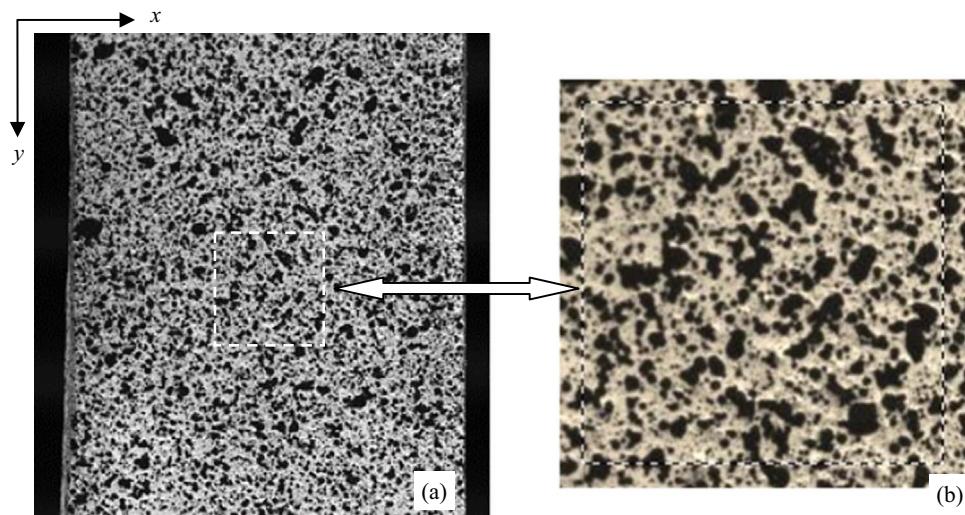


Fig. 5. (a) Undeformed speckle image and (b) rectangle ROI of the rubber specimen under tension.

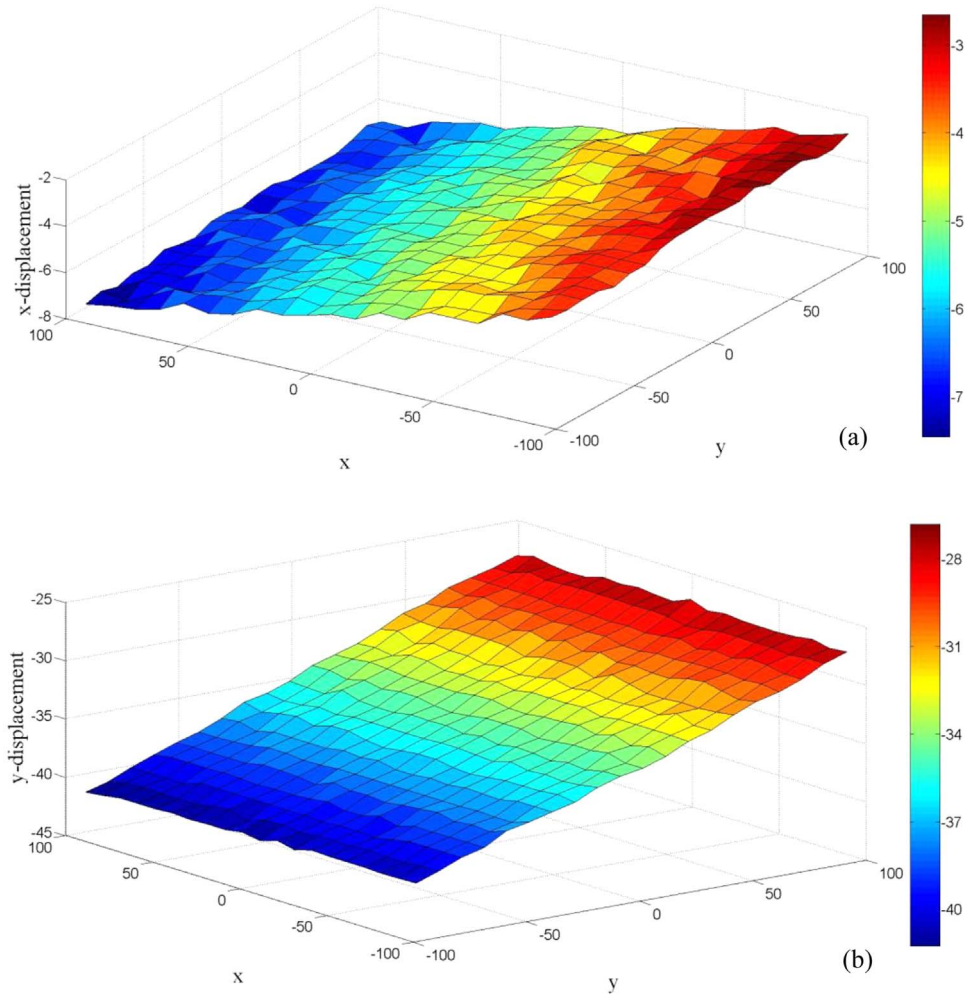


Fig. 6. (a) X-displacement and (b) y-displacement calculated by the proposed method for the rubber specimen under about 7.5% tension deformation.

DIC matching process for large deformation measurements. Theoretical error analysis proved that the new proposed second-order shape operator has stable second-order accuracy. In uniform large deformation measurement, it found through the numerical comparison of simulated speckle images that the proposed IC-GN2 algorithm had a higher accuracy than the FC-GN2 algorithm when the subset size was larger than 43 pixels. In non-uniform deformation simulation, the proposed new operator seems to have a relatively better performance than the previous IC-GN2. The rubber tensile experiment revealed that the FC-GN2 algorithm has the best performance under a large deformation of about 7.5%. When the large deformation was about 27%, the proposed IC-GN2 algorithm tended to match the precision of

the FC-GN2 algorithm and especially exceeded to the latter in precision when the subset size was larger than 43 pixels.

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Appendix A

In this section, a more rigorous form of the second-order shape operator is derived, which has a higher operator precision. To construct the reversible linear transformation matrix, we respectively introduce the displacement functions and their conjugate function as

$$\begin{cases} u^2(x, y) = x'^2(x, y) - x \\ v^2(x, y) = y'^2(x, y) - y \end{cases} \text{ and } \begin{cases} \bar{u}^2(x, y) = x'^2(x, y) + x \\ \bar{v}^2(x, y) = y'^2(x, y) + y \end{cases} \quad (\text{A1})$$

where (x, y) and (x', y') are the coordinates in the same point in the reference subset and target subset, respectively. Both x' and y' are functions of x and y , respectively. u and v are the displacements in the x -direction and y -direction, respectively. $\bar{u}$ and $\bar{v}$ are the conjugates of u and v , respectively. Obviously, x' and y' can be expressed in the following functions of x and y :

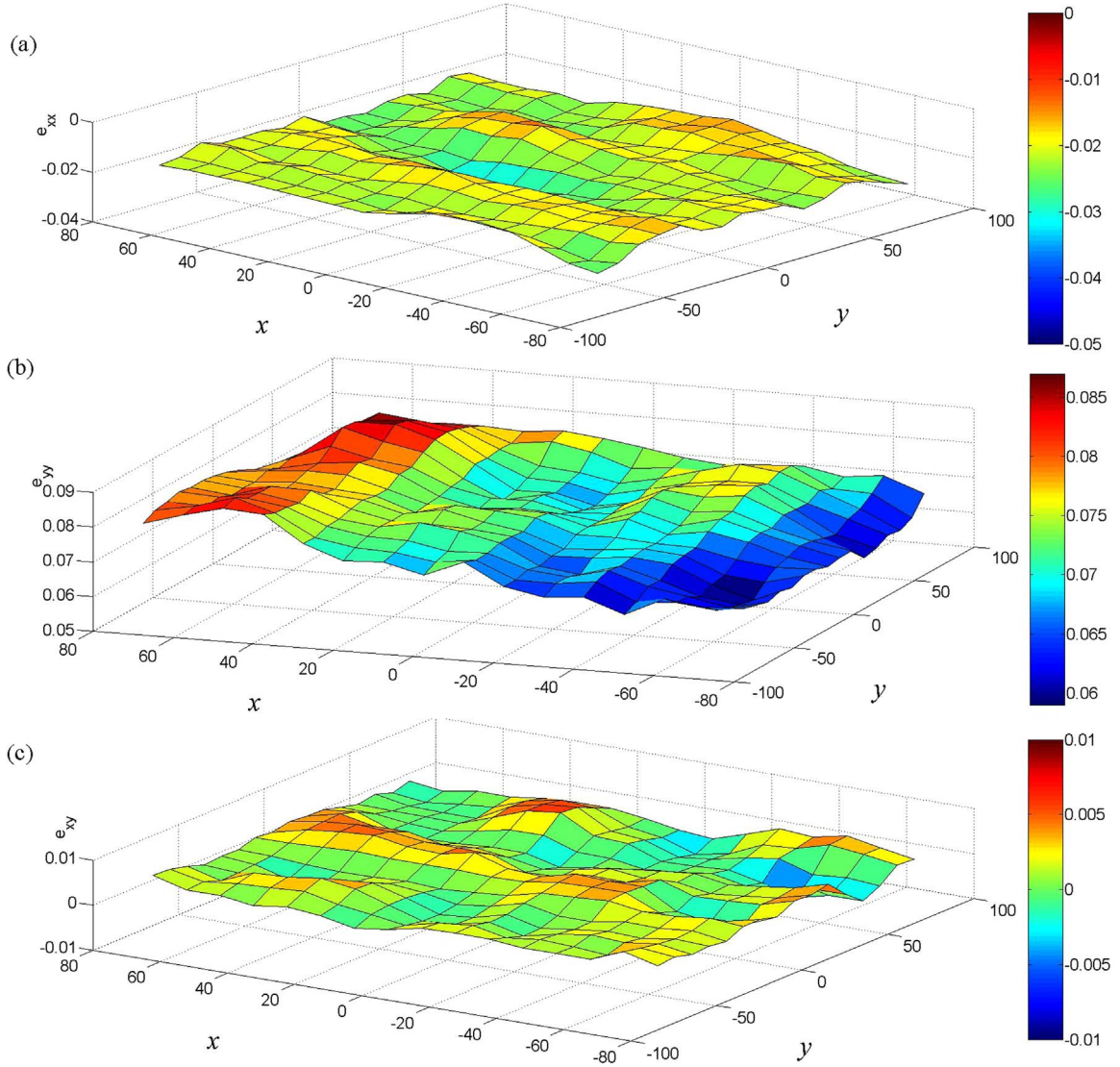


Fig. 7. (a) ε_{xx} (b) ε_{yy} and (c) ε_{xy} strains of the rubber specimen under about 7.5% tension deformation.

$$\begin{cases} x'^2 - x^2 = u\bar{u}(x, y) \\ x'y' - xy = \frac{1}{2}[u\bar{v}(x, y) + \bar{u}v(x, y)] \\ y'^2 - y^2 = v\bar{v}(x, y) \end{cases} \quad (\text{A2})$$

and the partial derivative relations of u and v are

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial x} + 2 \\ \frac{\partial u}{\partial y} = \frac{\partial u}{\partial y} \end{cases} \text{ and } \begin{cases} \frac{\partial v}{\partial x} = \frac{\partial v}{\partial x} \\ \frac{\partial v}{\partial y} = \frac{\partial v}{\partial y} + 2 \end{cases}. \quad (\text{A3})$$

Replacing with the partial derivatives of u and v instead of the partial derivations of $\bar{u}$ and $\bar{v}$ (Eq. (A3)), we can write the second-order Taylor expansion of the first right item in Eq. (A2) as

$$\begin{aligned} u\bar{u}(x, y) &= u\bar{u} + \left[\frac{\partial u\bar{u}}{\partial x} \frac{\partial u\bar{u}}{\partial y} \right] \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \frac{\partial^2 u\bar{u}}{\partial x^2} & \frac{\partial^2 u\bar{u}}{\partial x \partial y} \\ \frac{\partial^2 u\bar{u}}{\partial x \partial y} & \frac{\partial^2 u\bar{u}}{\partial y^2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + o(x^2 + y^2) \\ &= u\bar{u} + \left[\frac{\partial u}{\partial x} \bar{u} + u \frac{\partial \bar{u}}{\partial x} + \frac{\partial u}{\partial y} \bar{u} + u \frac{\partial \bar{u}}{\partial y} \right] \begin{bmatrix} x \\ y \end{bmatrix} \\ &+ \frac{1}{2} \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} \frac{\partial^2 u}{\partial x^2} \bar{u} + 2 \frac{\partial u}{\partial x} \frac{\partial \bar{u}}{\partial x} + u \frac{\partial^2 \bar{u}}{\partial x^2} & \frac{\partial^2 u}{\partial x \partial y} \bar{u} + \frac{\partial u}{\partial x} \frac{\partial \bar{u}}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial \bar{u}}{\partial x} + u \frac{\partial^2 \bar{u}}{\partial x \partial y} \\ \frac{\partial^2 u}{\partial x \partial y} \bar{u} + \frac{\partial u}{\partial x} \frac{\partial \bar{u}}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial \bar{u}}{\partial x} + u \frac{\partial^2 \bar{u}}{\partial x \partial y} & \frac{\partial^2 u}{\partial y^2} \bar{u} + 2 \frac{\partial u}{\partial y} \frac{\partial \bar{u}}{\partial y} + u \frac{\partial^2 \bar{u}}{\partial y^2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + o(x^2 + y^2). \end{aligned} \quad (\text{A4})$$

The above equation is further simplified as

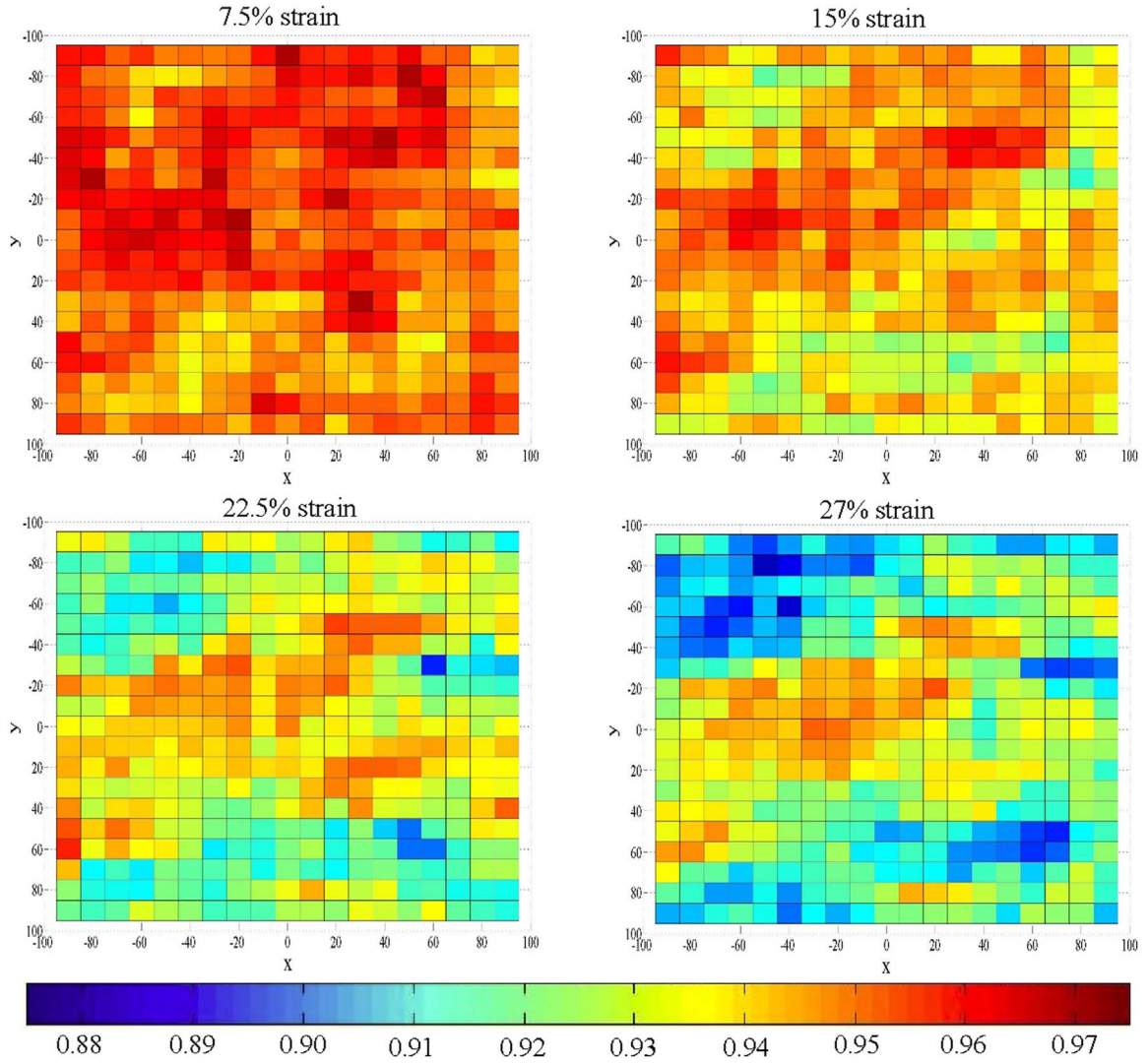


Fig. 8. Correlation coefficient distributions at different tension strains from 7.5% to 27%.

$$u\bar{u}(x, y) = u^2 + (4u + 2uu_x)x + 2uu_xy + (u_x^2 + uu_x + 4u_x)x^2 + 2(uu_{xy} + u_xu_y + 2u_y)xy + (u_y^2 + uu_{yy})y^2 + o(x^2 + y^2), \quad (A5)$$

where the subscripts are denoted as the first-order or second-order partial derivatives.

In the same way, the following equations of $v\bar{v}$, $u\bar{v}$ and $\bar{v}u$ are established as

$$v\bar{v}(x, y) = v^2 + 2vv_xx + 2(vv_y + 2v)y + (v_x^2 + vv_x)x^2 + 2(vv_{xy} + v_xv_y + 2v_x)xy + (v_y^2 + vv_{yy} + 4v_y)y^2 + o(x^2 + y^2), \quad (A6)$$

$$u\bar{v}(x, y) = uv + (u_xv + uv_x)x + (u_yv + uv_y + 4u)y + (\frac{1}{2}u_{xx}v + u_xv_x + \frac{1}{2}uv_{xx})x^2 + (u_{xy}v + uv_{xy} + u_xv_y + u_yv_x + 4u_x)xy + (\frac{1}{2}u_{yy}v + u_yv_y + \frac{1}{2}uv_{yy} + 4u_y)y^2 + o(x^2 + y^2), \quad (A7)$$

$$\bar{u}v(x, y) = uv + (u_xv + uv_x + 4v)x + (u_yv + uv_y)y + (\frac{1}{2}u_{xx}v + u_xv_x + \frac{1}{2}uv_{xx} + 4v_x)x^2 + (u_{xy}v + uv_{xy} + u_xv_y + u_yv_x + 4v_y)xy + (\frac{1}{2}u_{yy}v + u_yv_y + \frac{1}{2}uv_{yy})y^2 + o(x^2 + y^2). \quad (A8)$$

The following approximations are easily obtained as

$$x'^2 - x^2 = u\bar{u}(x, y) \approx u^2 + (4u + 2uu_x)x + 2uu_xy + (u_x^2 + uu_x + 4u_x)x^2 + 2(uu_{xy} + u_xu_y + 2u_y)xy + (u_y^2 + uu_{yy})y^2, \quad (A9)$$

$$y'^2 - y^2 = v\bar{v}(x, y) \approx v^2 + 2vv_xx + 2(vv_y + 2v)y + (v_x^2 + vv_x)x^2 + 2(vv_{xy} + v_xv_y + 2v_x)xy + (v_y^2 + vv_{yy} + 4v_y)y^2, \quad (A10)$$

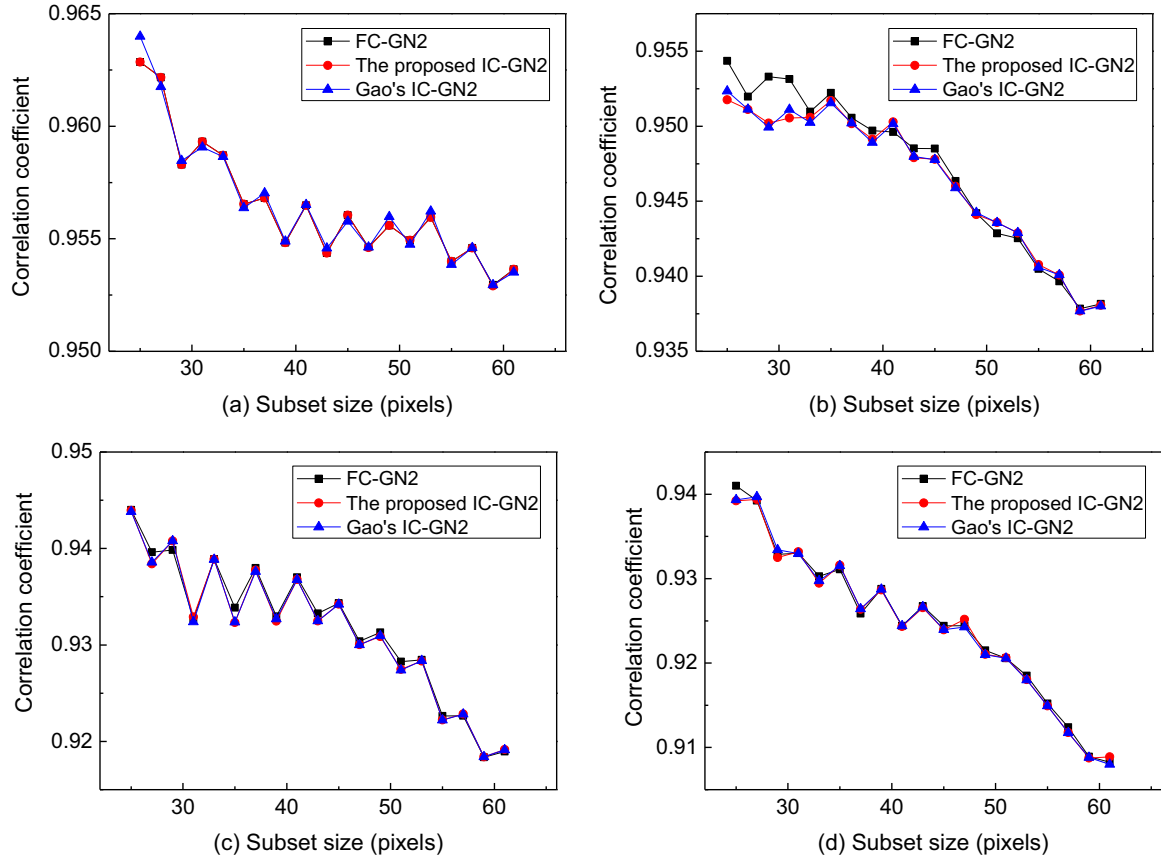


Fig. 9. Relationship between mean correlation coefficient and subset size under (a) 7.5% deformation, (b) 15% deformation, (c) 22.5% deformation and (d) 27% deformation.

$$\begin{aligned}
 x'y' - xy &= \frac{1}{2}[u\bar{v}(x, y) + \bar{v}v(x, y)] \\
 &\approx uv + (u_x v + uv_x + 2v)x + (u_y v + uv_y + 2u)y + (\frac{1}{2}u_{xx}v + u_x v_x + \frac{1}{2}u_{yy}v + 2v_y)x^2 \\
 &+ (u_{xy}v + uv_{xy} + u_x v_y + u_y v_x + 2u_x + 2v_y)xy + (\frac{1}{2}u_{yy}v + u_y v_y + \frac{1}{2}u_{xx}v + 2u_x)y^2.
 \end{aligned} \tag{A11}$$

Considering all of the above equations, we can construct a new complemented matrix in invertible linear transformation as

$$\mathbf{F}'(p)_{3 \times 6} = [\mathbf{C}'(p)_{3 \times 3} \quad \mathbf{D}'(p)_{3 \times 3}], \tag{A12}$$

where $p = [u \ u_x \ u_y \ u_{xx} \ u_{xy} \ u_{yy} \ v \ v_x \ v_y \ v_{xx} \ v_{xy} \ v_{yy}]^T$ is the second-order deformation parameter and the submatrices $\mathbf{C}'(p)$ and $\mathbf{D}'(p)$ have the following closed forms as

$$\mathbf{C}'(p) = \begin{bmatrix} u^2 & 2uu_x + 4u & 2uu_y \\ uv & u_x v + uv_x + 2v & u_y v + uv_y + 2u \\ v^2 & 2vv_x & 2vv_y + 4v \end{bmatrix}, \tag{A13}$$

$$\mathbf{D}'(p) = \begin{bmatrix} u_x^2 + uu_{xx} + 4u_x + 1 & 2uu_{xy} + 2u_x u_y + 4u_y & u_y^2 + uu_{yy} \\ \frac{1}{2}u_{xx}v + u_x v_x + \frac{1}{2}u_{yy}v + 2v_x & u_{xy}v + uv_{xy} + u_x v_y + u_y v_x + 2u_x + 2v_y + 1 & v_y^2 + vv_{yy} + 4v_y + 1 \\ v_x^2 + vv_{xx} & 2vv_{xy} + 2v_x v_y + 4v_x & \frac{1}{2}u_{yy}v + u_y v_y + \frac{1}{2}u_{xx}v + 2u_x \end{bmatrix}. \tag{A14}$$

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